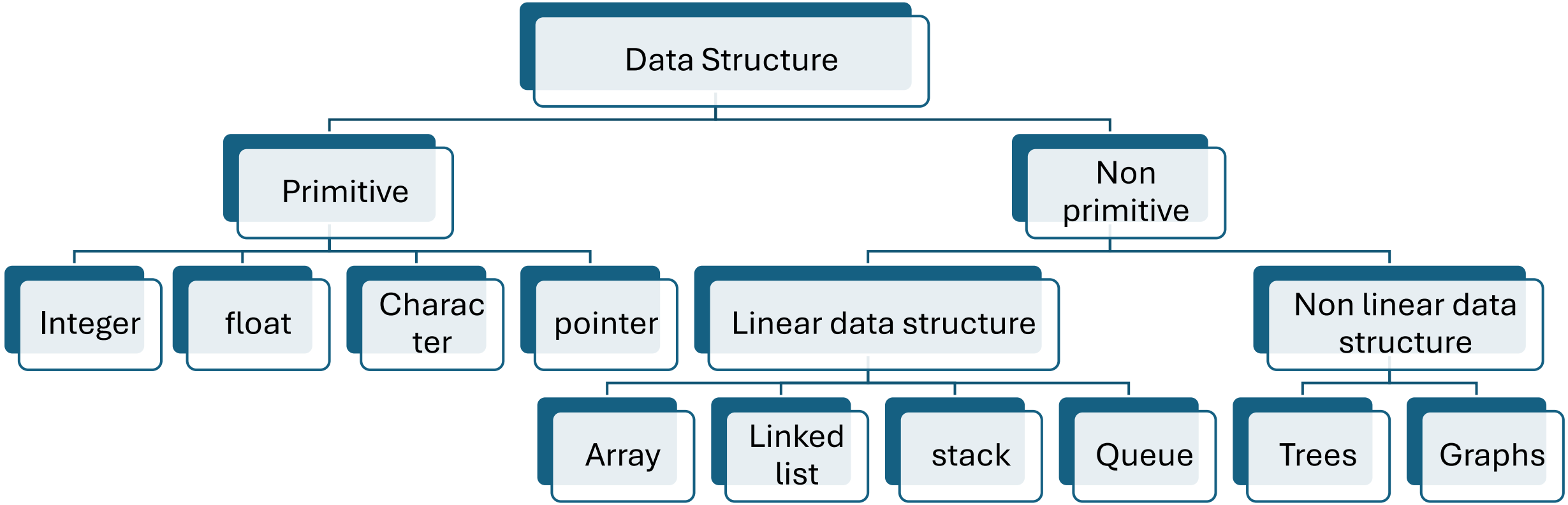
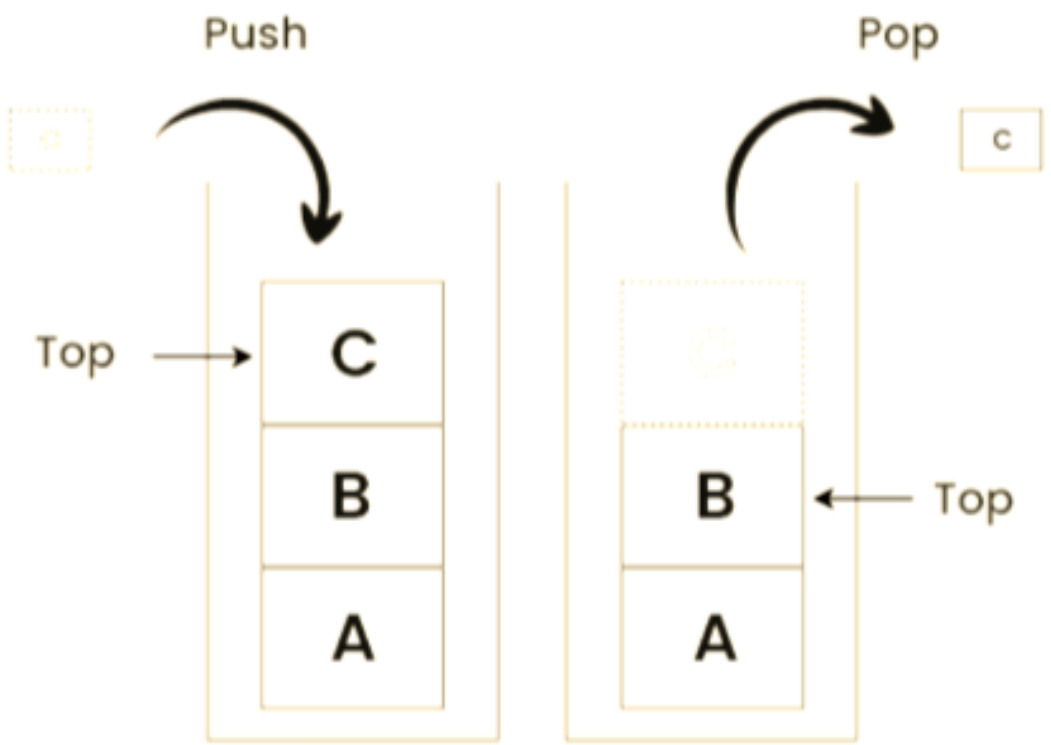

Data Structure
Primitive
Non primitive
Integer
float
Charac ter
pointer
Linear data structure
Non linear data structure
Array
Linked list
stack
Queue
Trees
Graphs

Stack
Data Structure

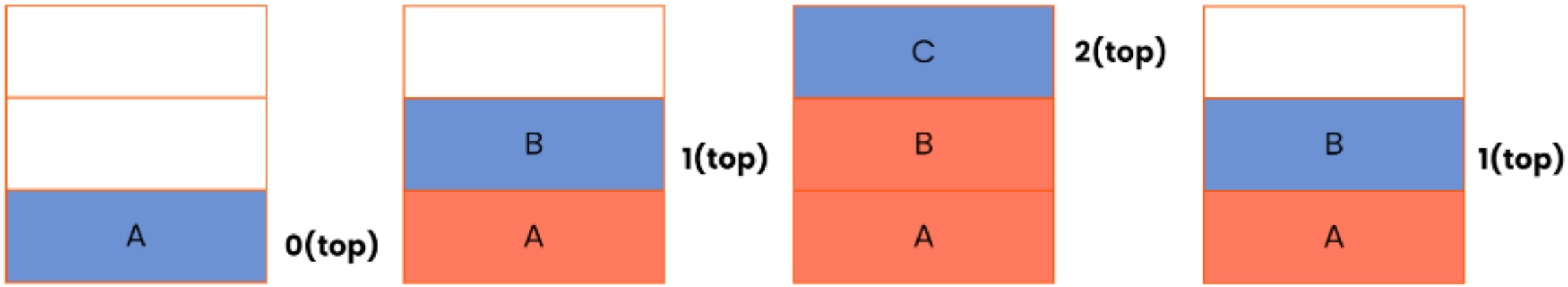
Stack is a linear data structure that follows a particular order in which the operations are performed. The order may be **LIFO(Last In First Out)** or **FILO(First In Last Out)**. **LIFO** implies that the element that is inserted last, comes out first and **FILO** implies that the element that is inserted first, comes out last.



Stack Array (with index)

- It is an array but has a special property that is last in first out (LIFO)

Size = 4



01

Example

Print stack [0]

A

02

Example

Print stack [top]

B

B
A

Example

Add(5)

Add(10)

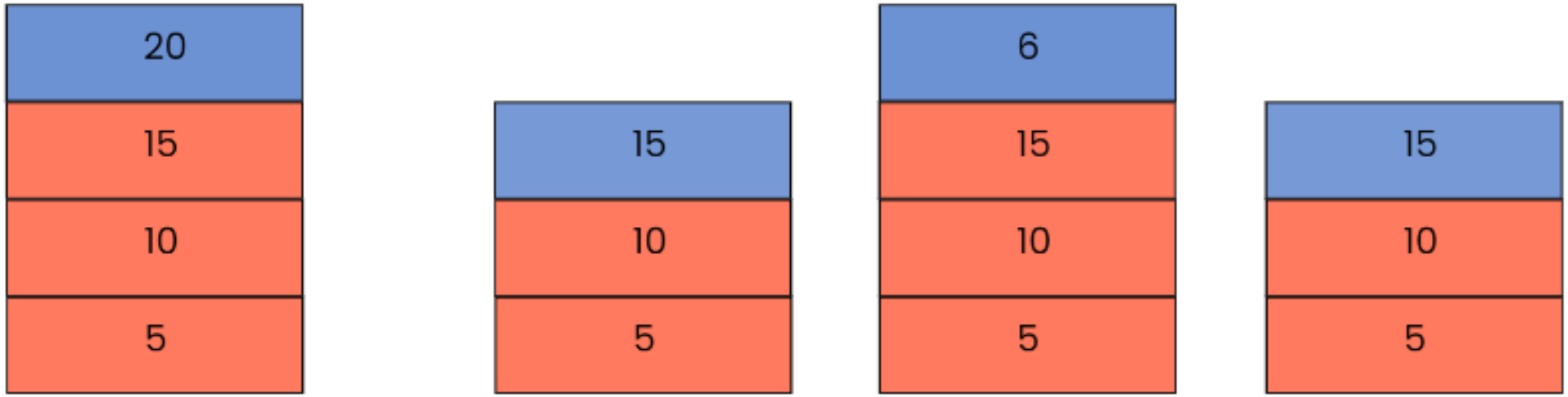
Add(15)

Add(20)

delete()

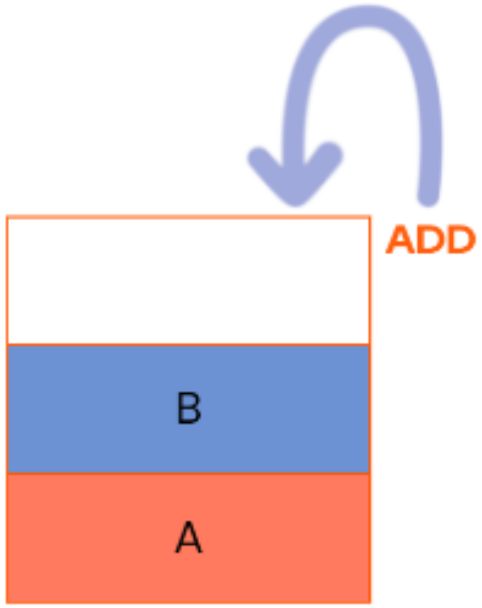
Add(6)

Delete()



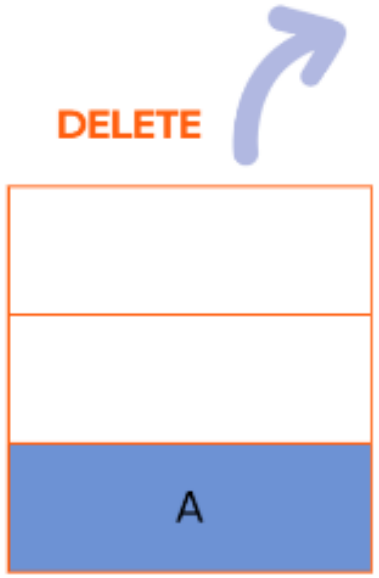
Common functions to build array stack

Display()
Print



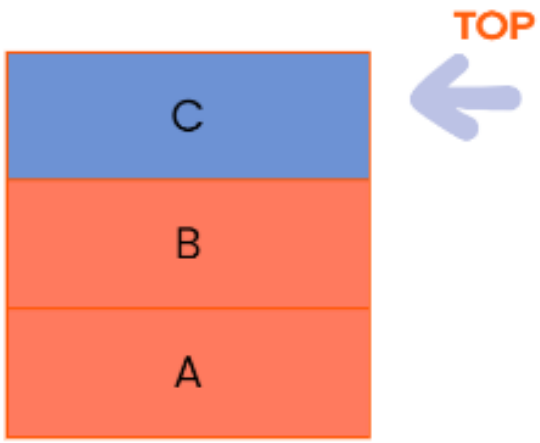
Push(val)

Adding



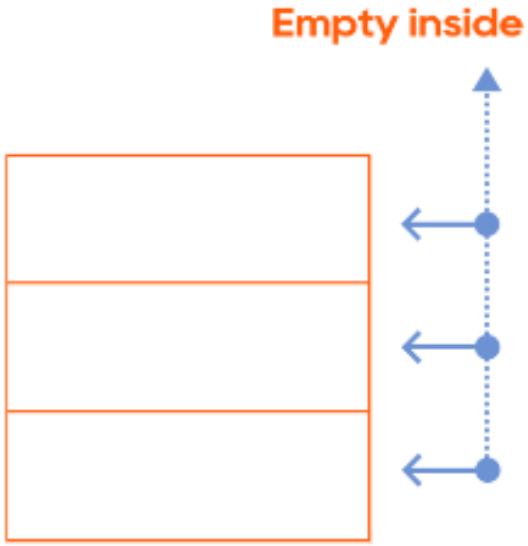
Pop()

deleting



getTop()

To find last element in stack



Is Empty()

Check the stack is empty or not

Disadvantages for stack array SIZE

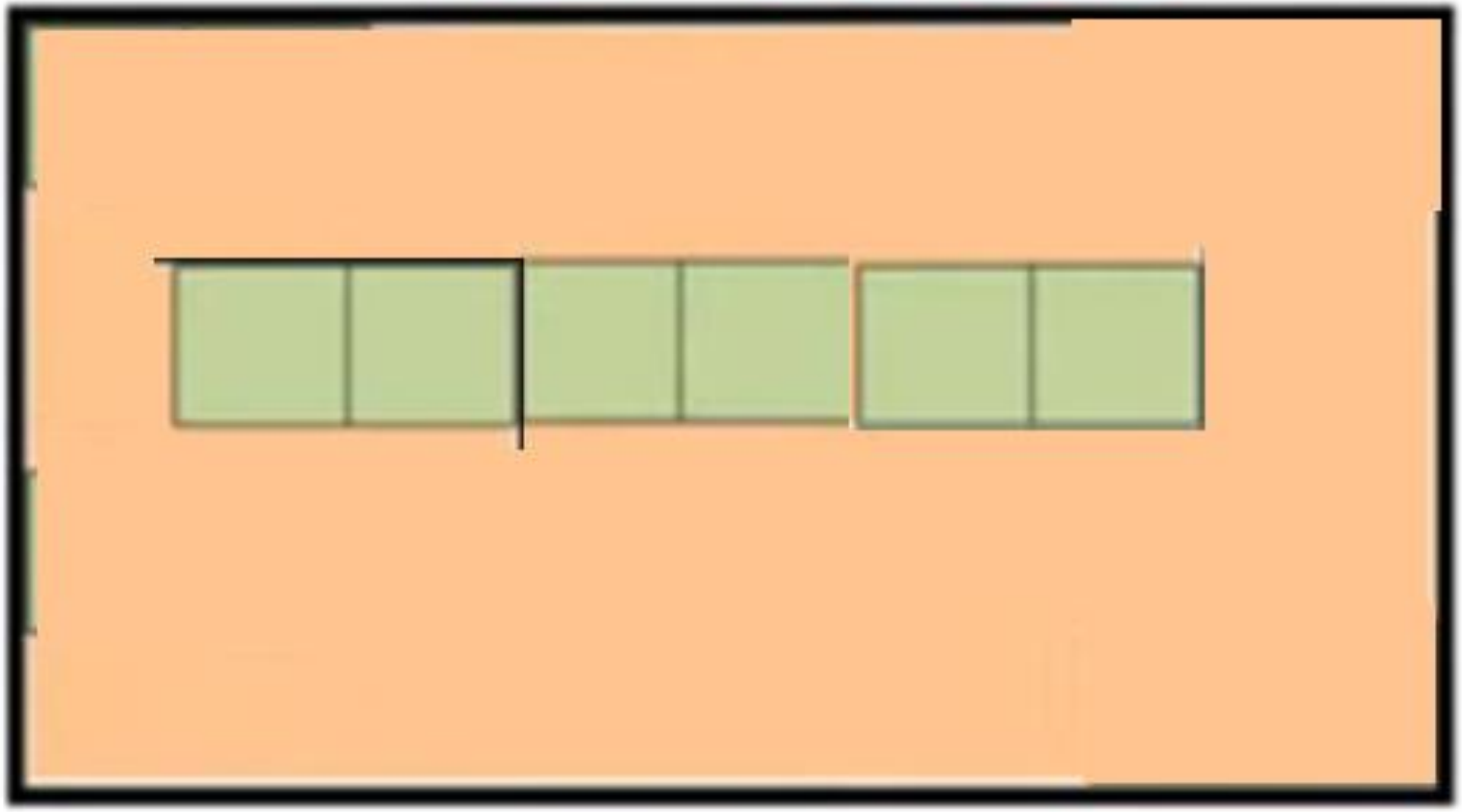
1- If I register 100 elements, and I find that I need more than that

0	1	2							99

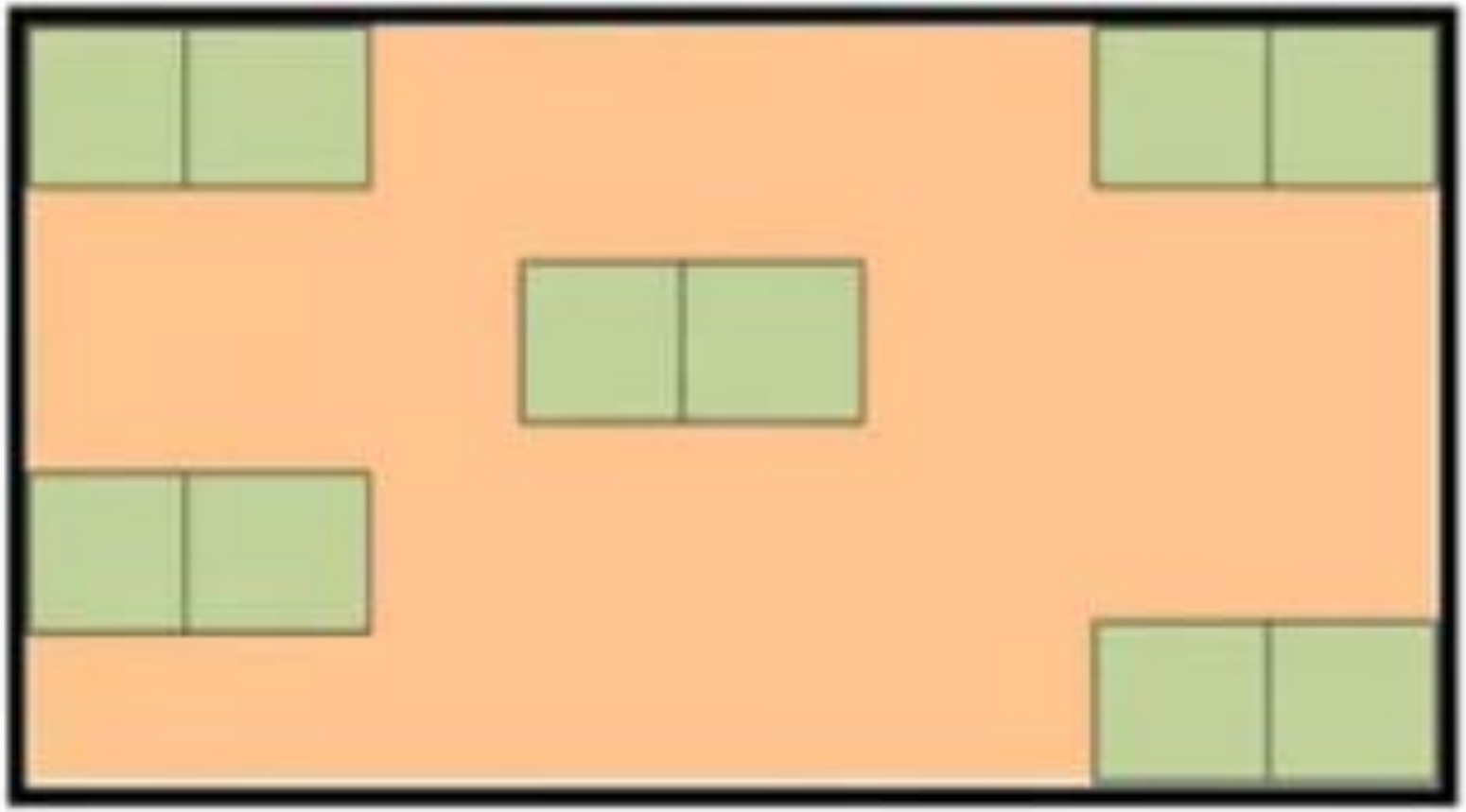
2- If I register 10000 elements, and I find that I don't need all of it

0	1								10000

MEMORY

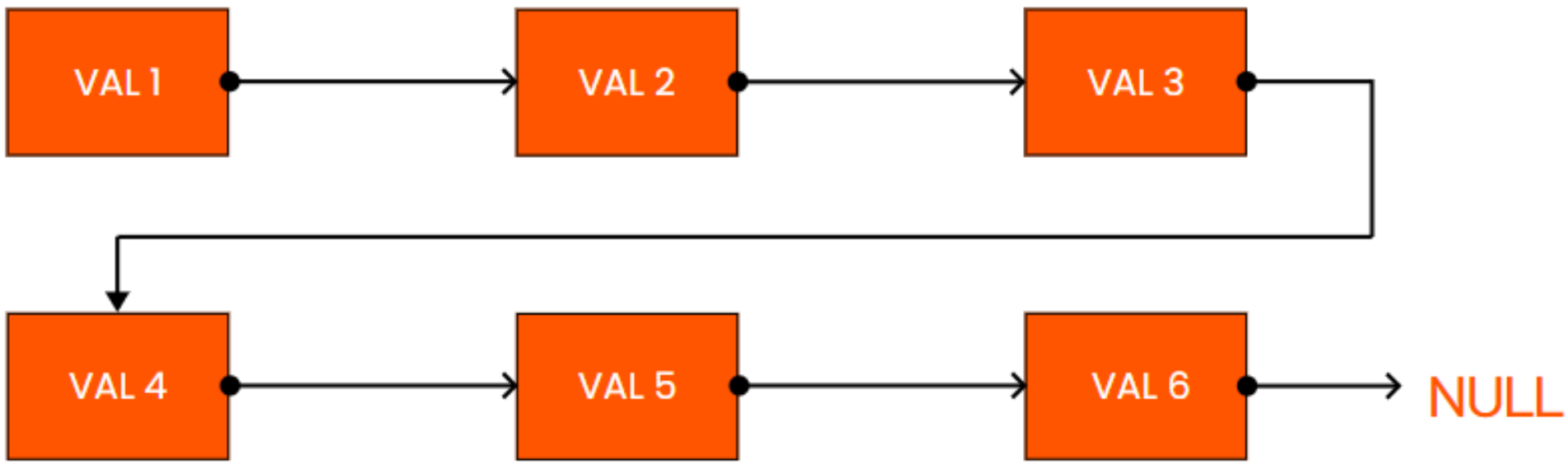


MEMORY



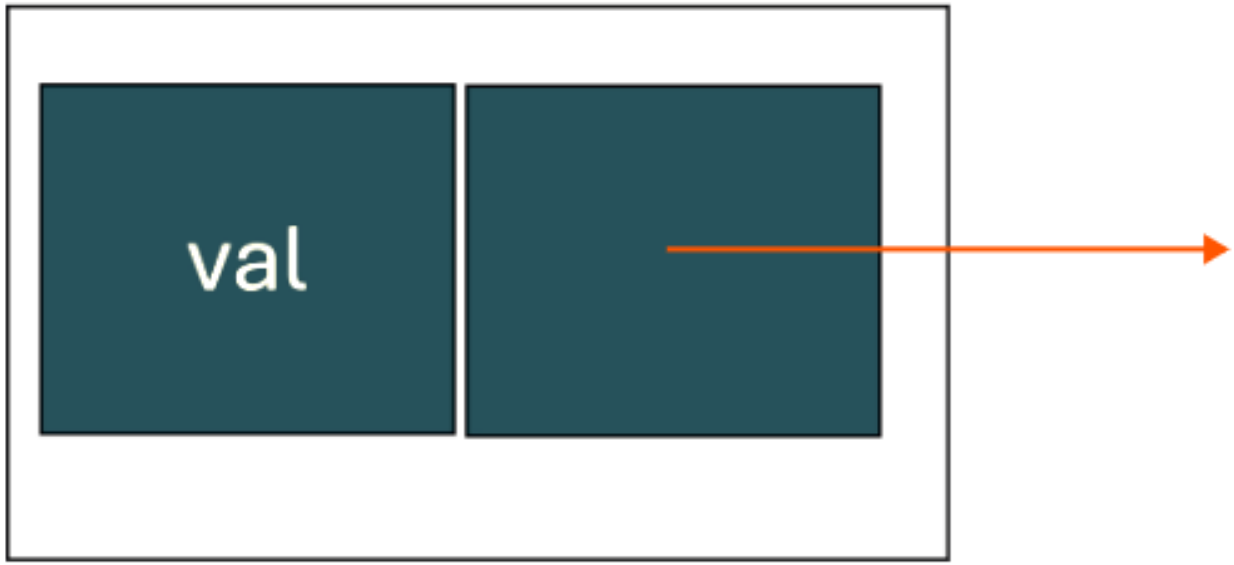
Stack using pointer

•That is a stack but doesn't need to allocate a specific size in the first, that depend on adding values in elements and each element has a pointer to the next element.

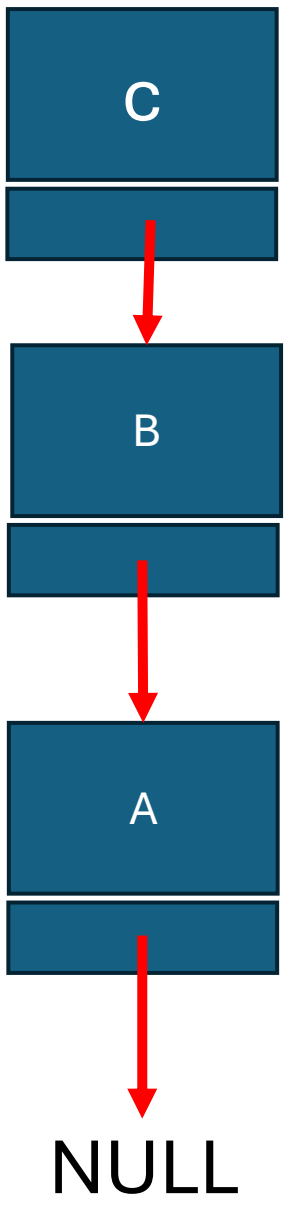


LINKED STACK

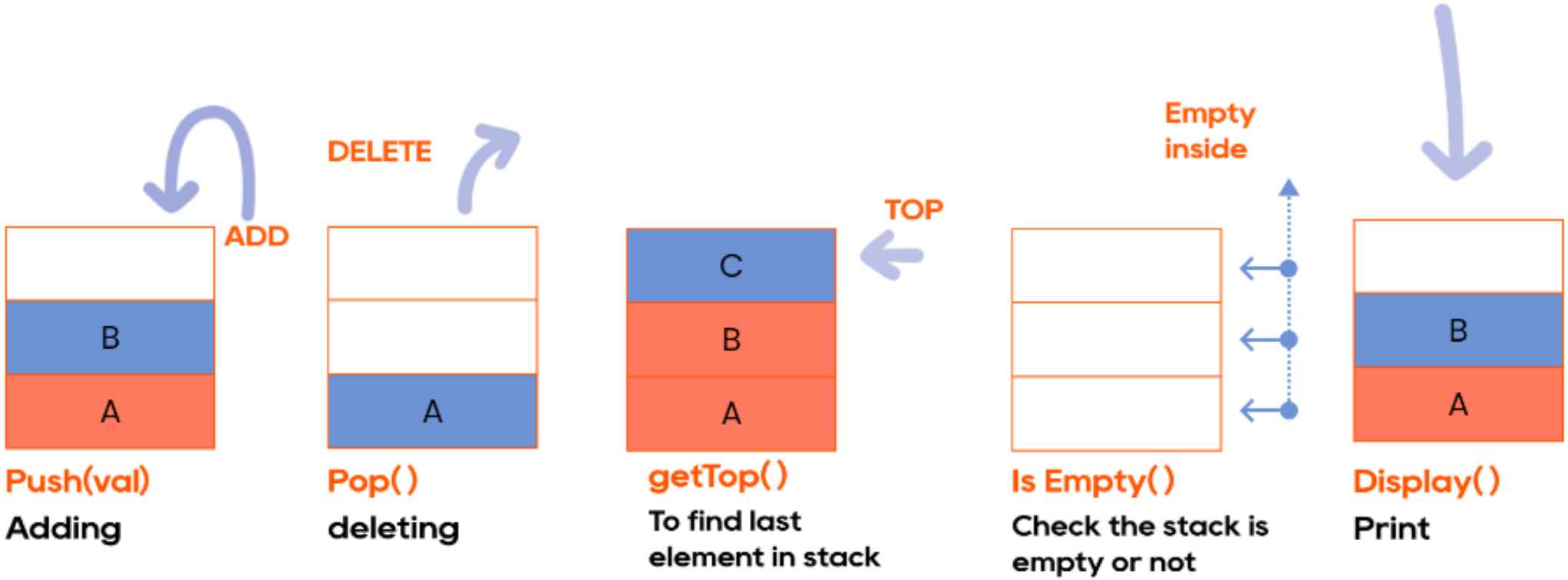
NODE (value + pointer)



Linked Stack



Common function to build Linked Stack



Application of Stack

Expressions evaluation

Expression Evaluation

- The priority of the operations

1. •Parenthesis
2. •The power
3. •Multiply / division
4. •Adding / subtraction

Example

3 * 5/ (52*4-3) +10 / 2

Sorting Operands and operations (3+5)

Infix 3 + 5

- **Postfix** 3 5 +
- **Prefix** + 3 5

I don't understand anything

Example 1

Infix 4 + 7 * 2

Postfix (4) + (7*2)

(4) +(7 2 *)

(4) (7 2 *) +

the final result is

4 7 2 * +

14

Arrange items
Postfix

4+14 = 18

Example 1

Infix 4 + 7 * 2

Postfix (4) + (7*2)

(4) +(7 2 *)

(4) (7 2 *) +

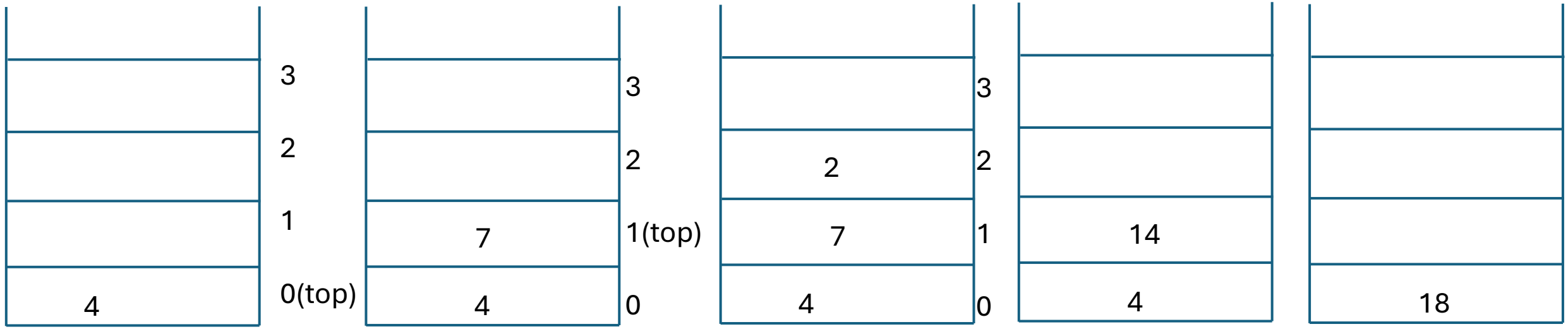
the final result is

4 7 2 * +

14

Arrange items
Postfix

4+14 = 18



Example 2

- Infix: A/B + 4
- Postfix: (A/B) + (4)

  (A B /) + (4)

  (A B /) (4) +

The final result : A B / 4 +

Example 3

- Infix: (5+2) * 3
- Postfix:

(5+2) * (3)

(5 2 +) (3) *

The final result : 5 2 + 3 *

Example 4

- Infix : (A*B) + (C/ D)
- Postfix:

(A*B) + (C/D)

(A B *) + (C D /)

(A B*) (C D /) +

The final result : A B * C D / +

Example 5

- Infix: 5/5+2-1*9
- Postfix:

(5/5)+(2)-(1*9)

(5 5 /) + (2) – (1 9 *)

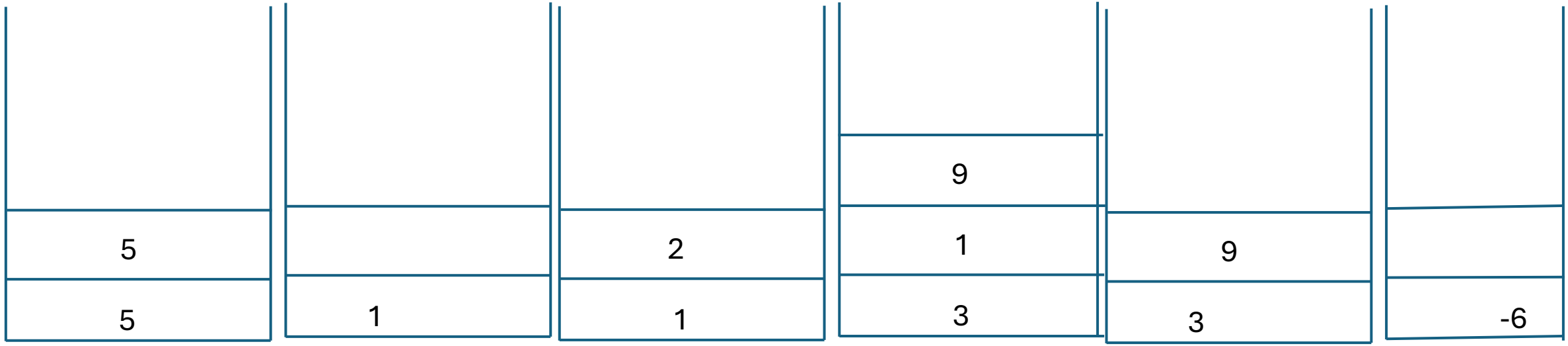
(5 5 /) (2) + (1 9 *) –

the final result

5 5 / 2 + 1 9 * -

How to use stack in the postfix
(5 5 / 2 + 1 9 * -)

- The stack work from lest to write → (5 5 / 2 + 1 9 * -)



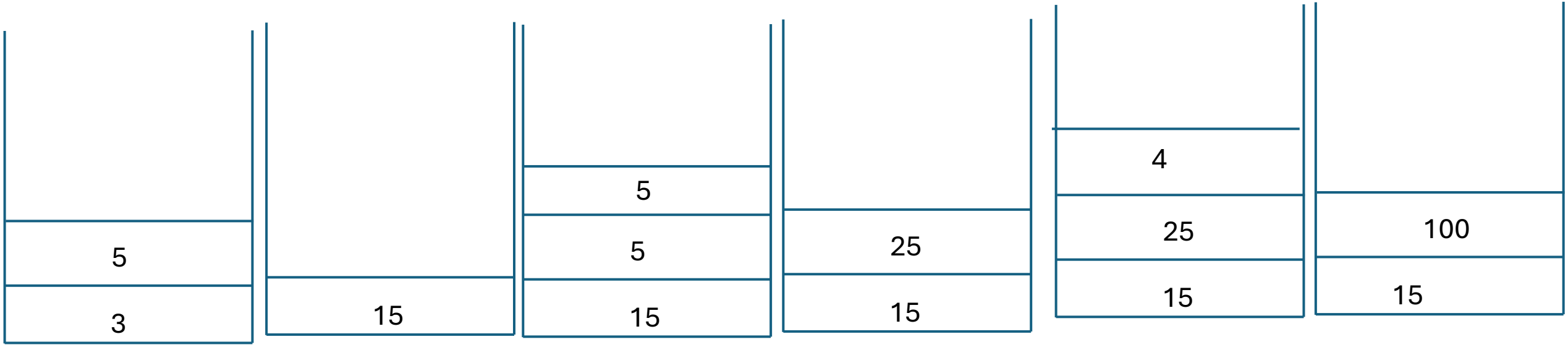
Example:

$$3 * 5/ (5^2*4-3) +10 / 2$$

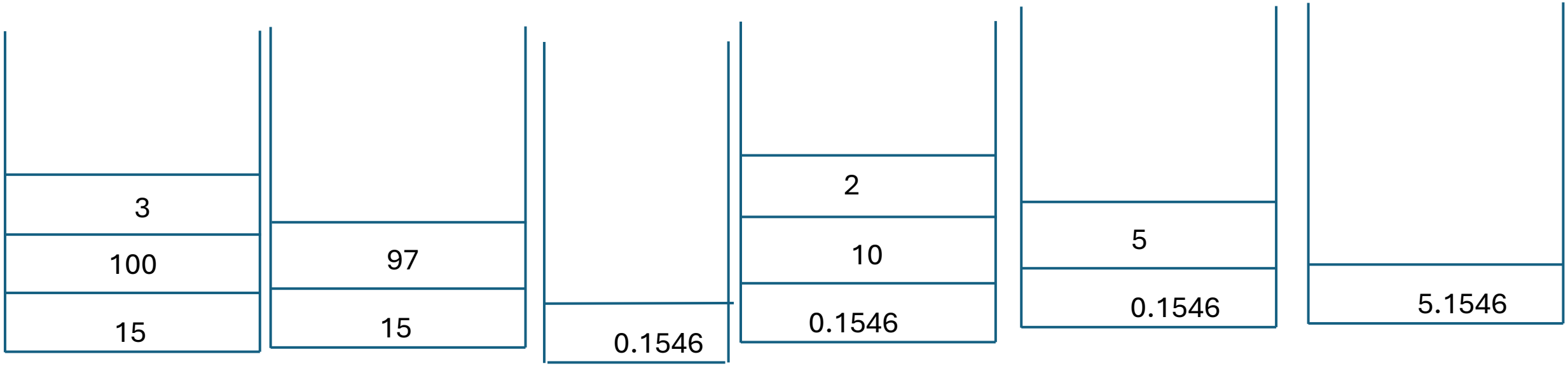
Example:

$$3 * 5/ (5^2*4-3) +10 / 2$$

- $(3*5)/ \ (((5^2)*4) – 3) + (10/2)$
- (3 5 *) /(((5*5) * 4)-3)+(10/2)
- (3 5 *) / (((5 5 *) *4) -3)+(10 2 /)
- (3 5 *) / ((5 5 * 4 *) -3) + (10 2 /)
- (3 5 *) / (5 5 * 4 * 3 -) + (10 2 /)
- 3 5 * 5 5 * 4 * 3 - / 10 2 / +


5
3
15
5
5
15
25
15
4
25
15
100
15

3 5 * 5 5 * 4 * 3 - / 10 2 / +


3
100
15
97
15
0.1546
2
10
0.1546
5
0.1546
5.1546

Postfix method

- Example: 4-6*5/8+2
- Solution: (4)- ((6*5)/(8)) +(2)
- = (4) – ((65*)/(8))+(2)
- =(4)-(65*8/)+(2)
- =465*8/-2+

- Start from left to the RIGHT in postfix.
- The less index is the first and high index is the second in subtraction and division.

5		8				
6	30	30	30/8=3.75		2	
4	4	4	4	0.25	0.25	2.25

The power in postfix

- Example:

8*5+3^2-4

= (8*5) + (3^2) – (4)

= (85*)+(32^)- (4)

= 85*32^+4-

		2				
5		3	9		4	
8	40	40	40	49	49	45

Prefix method

- 4-6*5/8+2

= (4)-((6*5)/8)+(2)

=(4)-((*65)/8)+(2)

=(4)-(/*658)+(2)

= +-4/*6582

- Start from **RIGHT** to the **LEFT** in postfix.
- The **HIGH** index is the **first** and **LESS** index is the **second** in subtraction and division.

6						
5	30			4		
8	8	30/8=3.75	3.75	3.75	0.25	
2	2	2	2	2	2	2.25

The power in prefix

- Example:

8*5+3^2-4

= (8*5) + (3^2) – (4)

= (*85)+(^32)- (4)

= -+*85^324

		8			
3		5	40		
2	9	9	9	49	
4	4	4	4	4	45

QUIZ

Q1- Find the complexity for the following

```
Int i;
i=1;
For (i=1; i<=n; i=i*2)
Print i;
For (int i=0; i<n; i++)
For (int j=0; j<n; j++)
For (int k=0; k<n; k=k*2)
Print i+j+k;
For (int i=0; i<n; i++)
For (int j=0; j<n; j=j/3)
Print i+j;
```

Q2- Find the comple for the following

```
int x=2;
i=1;
x=x+1;

for (i=1; i<=n; i++)  {
            //statement
}

for (i=1; i<=n; i++)  {
             for (j=1; j<=n; j++)  {
            //statement
            }
}

i=0;
For (i=0;i<n,i++)
Print I;
For(j=0;j<n; j=j/3)
Print j;
```

Solution Find the complexity for the following

```
Int i;
i=1;
For (i=1; i<=n; i=i*2)
Print i;
For (int i=0; i<n; i++)
For (int j=0; j<n; j++)
For (int k=0; k<n; k=k*2)
Print i+j+k;
For (int i=0; i<n; i++)
For (int j=0; j<n; j=j/3)
Print i+j;
```

$O(\log_2 n) + O(n^2 \log_2 n) + O(n \log_3 n)$

$$O(n^2 \log_2 n)$$

Solution- Find the complexity for the following

```
int x=2;
i=1;
x=x+1;

for (i=1; i<=n; i++)  {
            //statement
}

for (i=1; i<=n; i++)  {
             for (j=1; j<=n; j++)  {
            //statement
            }
}

i=0;
For (i=0;i<n,i++)
Print I;
For(j=0;j<n; j=j/3)
Print j;
```

Time complexity: $n^2+n+3 + 1 + n + \log n = O(n^2)$